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Nonnative Perception of Italian Consonant Length

Katherine Arnold
Department of Linguistics, University of California, Santa Cruz
kararnol@ucsc.edu

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Introduction

Geminates: long/double consonant

Singletons: short/single consonant

Short vowel Long consonant
f a tt o

f a t o
Long vowel Short consonant

L1 English speakers pay attention to the **vowel length** [4]

L1 Italian speakers pay attention to the **consonant length** [2]

- Studies of L2 length perception finds that **language proficiency plays an important role in native like** perception [1][3][5]

Research Question

- Do English speakers with more Italian proficiency pay attention to consonant length more than vowel length?
- How will length labelling vary across participant groups?
- How does voicing impact length labeling?

Methods

- Three groups of UCSC students with various Italian proficiencies:
  - Naïve (n=64) had no experience learning Italian
  - Beginner (n=12) within their first year of taking Italian
  - Advanced (n=13) enrolled in second year Italian courses, had studied abroad, or taken upper division language courses.

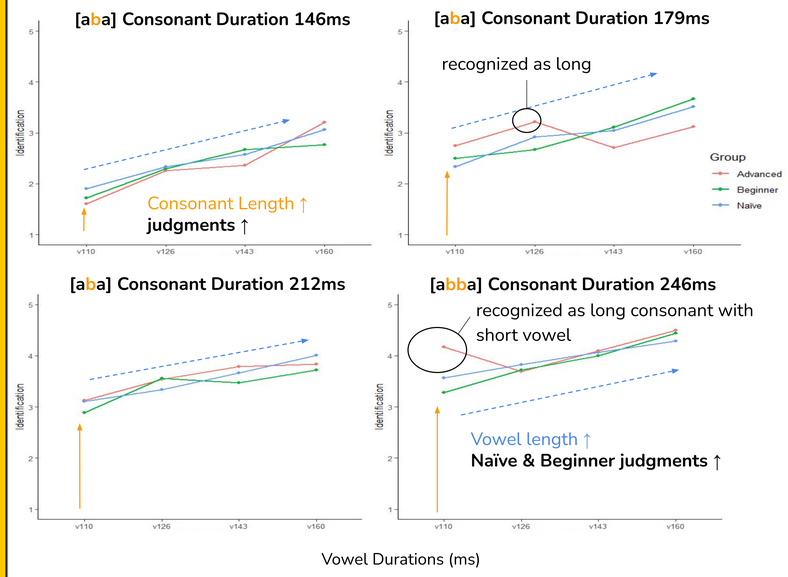
Stimuli

- The **fake and low frequency words** [apa], [appa], [aba], and [abba] were recorded by an Italian native speaker
- The durations of the first vowel and the intervocalic consonant were manipulated in Praat to create **4 synthetic durations** for each condition
  - 32 stimuli in total

Procedure

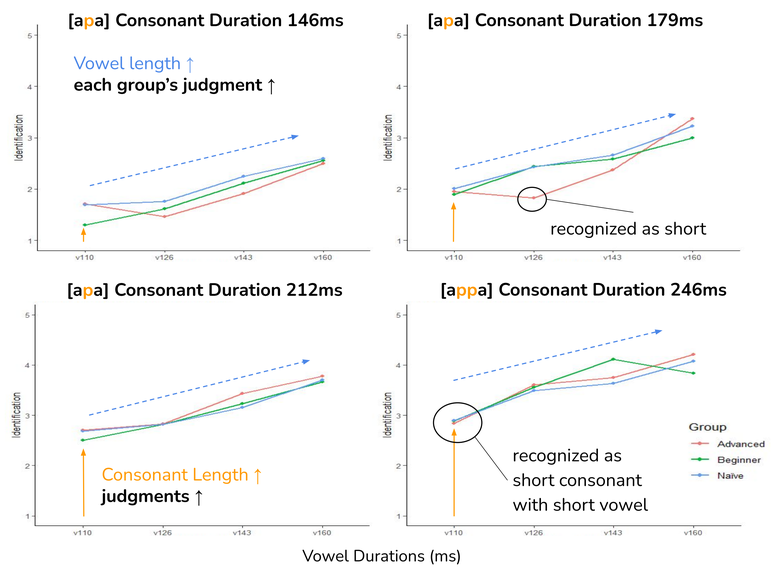
- Identification task
- Each sound played through headphones in a randomized order
- Participants **rank** the sound by length on a **scale of 1-5** (short = 1, long =5)

Voiced



- Naïve and Beginner groups identified length similarly
- Advanced group able to accurately identify geminate

Voiceless



- [apa] perceived as shorter than [aba]
- Advanced group unable to identify geminate

Results

- Mean rankings increased as vowel and consonant length increased
  - Each participant group is sensitive to consonant duration
- As Italian proficiency increased, participants relied less on vowel duration
  - Naïve participants relied on vowel and consonant duration roughly the **same amount**
  - Beginner group also relied on vowel and consonant duration
    - Their rankings did not have the same degree of positive linearity as the Naïve group
  - Advanced group relied more on **consonant duration** than on vowel duration
    - the two intermediate vowel steps (v126, v143) did not have a significant effect on the Advanced group's average identifications
- Voiced condition perceived as **longer** than voiceless
  - English listeners are sensitive to vowel length and use it to detect voicing [4]

Main Takeaways

- Language proficiency has a positive impact on length perception
- Voicing impacts perception of length at all levels of proficiency
  - not attested in the previous literature

Future Directions

- How does consonant voicing impact length perception in other places and manners of articulation?
- How would the study differ with the inclusion of real words?

Selected References: **[1]**De Clercq, B., Simon, E., & Crocco, C. (2014). Rosa versus rossa: The acquisition of Italian geminates by native speakers of Dutch. Phrasis: Studies in Language and Literature, 2(2), 3-29. **[2]** Esposito, A., & Di Benedetto, M. G. (1999). Acoustical and perceptual study of gemination in Italian stops. The Journal of the Acoustical Society of America, 106(4), 2051-2062. **[3]** Feng, Q., & Busà, M. G. The Classification of Native and Non-Native Productions of Italian Consonant Length Contrast. **[4]** Lisker, L. (1957). Closure Duration and the Intervocalic Voiced-Voiceless Distinction in English. Language, 33(1), 42–49. https://doi.org/10.2307/410949 **[5]** Tsukada, K., & Hajek, J. (2019). Misperception of Italian singleton and geminate obstruents by native Japanese speakers. Journal of the Phonetic Society of Japan, 23, 148-156.

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